

DISCRETE DISTRIBUTIONS:

BERNOULLI DISTRIBUTION:

• SCENARIO: A single trial (Bernoulli Trial) with two possible outcomes: 1 $\rightarrow$ Success, 0 $\rightarrow$ Failure.

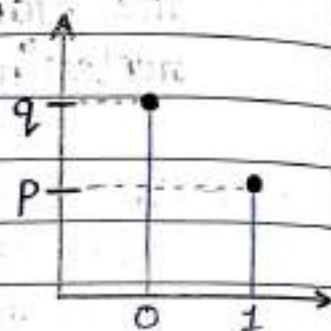
• P.M.F for Bernoulli Distribution:

• $p \in$ PROBABILITY OF SUCCESS

• $q \in$ PROBABILITY OF FAILURE

$$P.M.F = \begin{cases} p, & \text{if } x=1 \\ q=1-p, & \text{if } x=0 \end{cases}$$

$$f(x, p) = p^x (1-p)^{1-x}, \text{ where } x \in \{0, 1\}$$



• MEAN = p

• VARIANCE = $p \times q = p(1-p)$

BINOMIAL DISTRIBUTION:

• Binomial distribution tells you the probability of getting a certain number of successes in a fixed number of independent trials, where each trial has only two outcomes: success or failure.

(1)

(0)

• BINOMIAL DISTRIBUTION answers this question:

"If I repeat the same experiment many times where each time is success or failure, what is chance of getting 0, 1, 2, ... successes?"

• WHEN IT APPLIES ?

Use Binomial when :-

- You do something a fixed number of times : n
- Each try has only two outcomes : success or failure
(Head/Tail, correct/wrong, defect/no defect)
- The chance of success is the same every time : p
- Each try is independent (one result does not change the next).

Then the random variable, $X = \text{"number of successes in } n \text{ trials"}$
Follows a binomial distribution :

$$X \sim \text{Bin}(n, p)$$

• FORMULA :-

* Probability of getting k successes :-

$$P(X=k) = {}^nC_k p^k q^{n-k}$$

- p = Probability of success
- $q = (1-p)$ = the probability of failure
- n = the number of trials / experiments
- p^k = probability of those k successes

$$\rightarrow {}^nC_k = \left(\frac{n!}{k! \cdot (n-k)!} \right)$$

⇒ For Example :- A fair coin is tossed 3 times.

• Taking "Head" as success, so $p = 0.5$

• let $X =$ number of heads in 3 tosses

Then $X \sim \text{Bin}(3, 0.5) \approx \text{Bin}(n, p)$

• $P(X=0)$: no head at all

• $P(X=1)$: exactly one head

• $P(X=2)$: exactly two heads

or $P(X=3)$: exactly three heads

$$\text{let say } P(X=2) = {}^3C_2 (0.5)^2 (0.5)^1 = \frac{3!}{2!(3-2)!} \times 0.25 \times 0.5$$

$$= 3 \times 0.25 \times 0.5$$

$$= 0.375$$

Make a small table of $k=0,1,2,3$ and all $P(X=k)$, or

$$P(X=k) = {}^nC_k p^k q^{n-k} = (n-k)q$$

* NOTE :- MEAN = np

$$\text{VARIANCE} = npq$$

$$= np(1-p)$$

Example : Customer Conversion Rate

Q. Imagine you are an e-commerce business owner, and you want to analyze the conversion rate of visitors to your website. In this scenario, conversion means making a purchase. You're interested in making & understanding the likelihood of a specific number of purchases (success) out of a fixed number of website visits (trials) during a given period.

What is the probability of exactly 75 visitors making a purchase out of 100 website visits, given a 15% conversion rate.

Sol. n (number of website visits) = 100
 p (probability of making purchase on each visit) = 15% C-rate
 $= 0.15$
 x (number of purchases) = 75

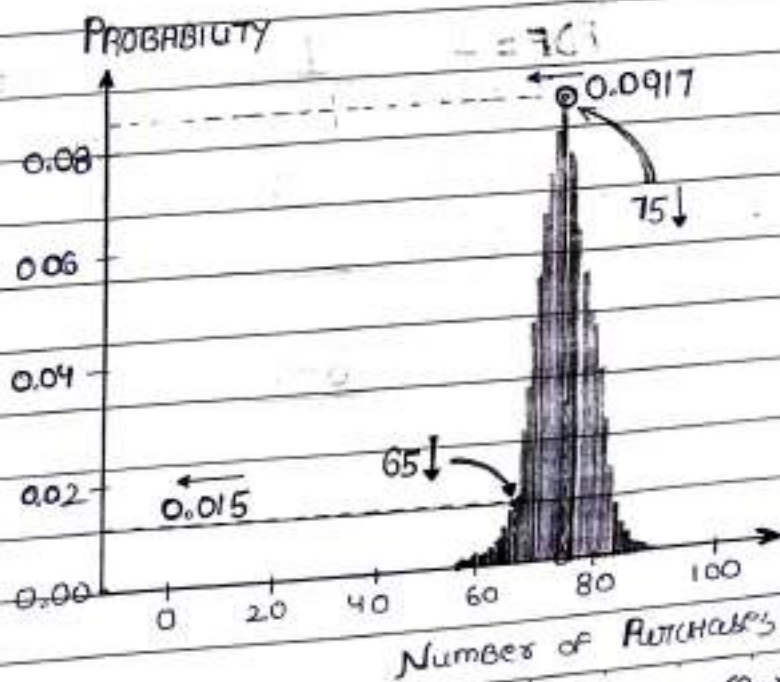
* We know: $PMF = \frac{100!}{(100-75)! \times 75!} \times 0.15^{75} \times (1-0.15)^{(100-75)}$

$$= \frac{100!}{25! \times 75!} \times (0.15)^{75} \times (1-0.15)^{25}$$

$$= 0.0917$$

$$= 9.17\%$$

⇒ तो 9.17% PROBABILITY है कि 100 में से 75 Customer जो website visit (trial) करते हैं वो purchase (success) करेंगे।



• Similarly we can change 'x' to get $P(x)$